Notes: Oyer (QJE, 1998)

Fiscal Year Ends and Nonlinear Incentive Contracts: The Effect on Business Seasonality

Contents

1 Big picture

- **Question.** Do nonlinear incentive contracts for salespeople and executives create fiscal-year seasonality in firm revenues and prices?
- **Core idea.** Bonus plans and sales quotas are commonly measured over the fiscal year. Agents then have incentives to pull sales into the current fiscal year, push sales into the next fiscal year, vary effort, and adjust prices around quota deadlines.
- **Empirical strategy.** The paper exploits variation in firms' fiscal year ends. Because firms end fiscal years in different calendar months, the paper can separate ordinary calendar seasonality from fiscal-year seasonality.
- **Headline result.** Manufacturing sales are lower in the first fiscal quarter and higher in the fourth fiscal quarter, conditional on calendar seasonality.
- **Price result.** The first fiscal quarter shows higher price growth and the fourth fiscal quarter shows lower price growth. This is consistent with low prices or high effort being used to bring sales forward at year end, followed by early-year price recovery.
- **Heterogeneity result.** Fiscal-year revenue effects are stronger in industries where salespeople can directly influence customer timing, especially wholesale and direct-sales channels, and where salesperson turnover is high.
- **Alternative explanations.** The paper checks whether fiscal years are merely chosen to match demand seasonality. Results survive alternative calendar controls and are supported by exogenous fiscal-year changes after mergers.
- **Takeaway.** Incentive contracts can generate real seasonality in sales and prices. The timing of accounting and compensation periods is an economic variable, not a neutral reporting convention.

2 Incremental contribution of the paper

- **From accounting manipulation to real business timing.** Earlier work, such as Healy-style bonus-plan studies, emphasized accrual manipulation. Oyer studies real sales timing and price movements caused by incentives.
- **A fiscal-year identification design.** The paper uses cross-firm variation in fiscal year ends to difference out calendar business cycles and isolate fiscal-year incentives.
- **Micro incentives with macro seasonality.** The paper shows how salesperson and executive compensation can affect aggregate-looking seasonality in manufacturing sales, connecting contract theory to business-cycle measurement.
- **Model of timing gaming.** It develops a simple dynamic model in which agents decide whether to pull sales into the current period or push sales out, depending on quota incentives and future costs.
- **Prices as a diagnostic.** By examining price movements as well as revenue movements, the paper distinguishes pure demand timing from incentive-induced pricing and effort responses.
- **Industry heterogeneity evidence.** It connects fiscal effects to industry characteristics: durability, customer type, sales channel, salesperson turnover, education, and earnings.
- **Causal discipline through fiscal-year changes.** The paper uses merger-induced fiscal-year changes and rare voluntary fiscal-year changes to test whether fiscal year timing causes revenue seasonality.

Interpretation: The incremental contribution is to demonstrate that nonlinear incentives can alter the timing of actual transactions, not merely reported accounting numbers. The paper turns fiscal-year-end variation into a way to observe agency problems in real product markets.

3 Data and measurement

3.1 Compustat manufacturing sample

The main sample uses Standard and Poor's Compustat quarterly industrial files for manufacturing firms. The empirical unit is the firm-quarter. Revenue is measured as quarterly sales, and prices are approximated using the change in log sales minus the change in log cost of goods sold.

- The core revenue outcome is the change in log fiscal-quarter sales from the previous fiscal quarter.
- The price outcome is the change in $\log(sales) - \log(cost\ of\ goods\ sold)$ from the previous quarter.
- Fiscal-quarter indicators identify whether the observation is the first or fourth fiscal quarter.
- Calendar-month effects absorb ordinary calendar seasonality.

Interpretation: The key measurement idea is to compare firms with different fiscal year ends but similar calendar environments. This lets the paper separate a December holiday effect from a fourth-fiscal-quarter incentive effect.

3.2 Industry-level sample restrictions

The industry regressions require that industries have enough variation in fiscal year ends and enough observations. Table I documents the distribution of fiscal year ends in selected industries.

- Most firms use December fiscal year ends, but many firms do not.
- Variation across fiscal year end months is essential for identification.
- Industry-level analyses are reported for two-, three-, and four-digit classifications where the sample has enough firms and fiscal-year-end variation.

Interpretation: If almost every firm in an industry had the same fiscal year end, one could not distinguish fiscal seasonality from industry calendar seasonality. The paper therefore needs cross-firm variation in fiscal year timing.

3.3 Sales force and industry characteristics

To interpret heterogeneity, the paper links fiscal-quarter effects to industry characteristics:

- whether the product is durable;
- whether sales are direct, wholesale, retail, or through other channels;
- salesperson turnover from SIPP data;
- education and earnings for salespeople from Census data.

Interpretation: These variables test the mechanism. Timing gaming should be more plausible when salespeople interact directly with customers, face turnover risk, or sell goods with negotiable timing.

4 Models and empirical specifications

4.1 Salesperson/executive incentive logic

The paper studies nonlinear contracts such as quotas and bonus thresholds. The stylized compensation schedule has low marginal pay before the threshold, high marginal pay around the quota/threshold, and potentially lower marginal pay after the target has been met.

$$Pay = f(Sales_{FY}), \qquad f''(\cdot) \neq 0 \text{ around quota or threshold.}$$

Interpretation: Nonlinearity creates timing incentives. A salesperson close to quota has high marginal value from moving one more sale into the current fiscal year, while a salesperson far below or far above quota may have different incentives.

4.2 Timing-gaming model

The two-period model has sales that could occur naturally in one period but can be pulled forward or pushed backward. Let x be first-period sales, Q^* be a quota or target, and B and C denote marginal benefits and costs of timing manipulation.

$$\text{Pull-in if } B_y + B_{z+} > C_{pull}; \qquad \text{Push-out if } B_{z-} > C_{push}.$$

- Pull-in means moving future sales into the current fiscal year.
- Push-out means delaying sales into the next fiscal year.
- The relative size of the pull-in and push-out regions depends on quotas, marginal compensation, future costs, and discounting.

Interpretation: The model predicts higher fourth-fiscal-quarter sales for firms with nonlinear incentive contracts. It also predicts lower first-fiscal-quarter sales because some sales have already been pulled forward into the prior year end.

4.3 Main revenue regression

The main revenue equation is a firm-quarter panel regression:

$$\Delta \log Sales_{i,q} = \alpha_i + \sum_m \gamma_{industry,m} CalendarMonth_{m,q} + \beta_1 \Delta 1\{FQ = 1\}_{i,q} + \beta_4 \Delta 1\{FQ = 4\}_{i,q} + \varepsilon_{i,q}.$$

The industry version interacts first- and fourth-fiscal-quarter indicators with three-digit industry dummies.

Interpretation: The coefficients compare changes in fiscal-quarter status within firms while flexibly controlling for calendar seasonality. A negative first-quarter coefficient and positive fourth-quarter coefficient are the core fiscal-year effect.

4.4 Price regression

The price proxy is:

$$\Delta \left[\log Sales_{i,q} - \log COGS_{i,q}\right].$$

The regression structure mirrors the revenue specification.

Interpretation: If firms or salespeople cut prices at the end of the fiscal year to pull sales forward, price changes should be lower in the fourth fiscal quarter and higher in the first fiscal quarter. This is the opposite of the revenue pattern.

4.5 Second-stage industry regressions

Let $f_{j,q}$ be the estimated fiscal-quarter effect for industry j and fiscal quarter q. The paper regresses these estimated industry effects on industry characteristics:

$$\widehat{f}_{j,q} = \theta X_j + u_j.$$

Interpretation: These regressions ask whether industries with more direct customer contact, turnover, durability, or salesperson characteristics have stronger fiscal-year effects.

5 Identification and empirical design

5.1 Core identification

The identifying variation is that fiscal years end in different calendar months across firms. Calendar business cycles affect all firms in a product market at the same calendar time, while incentive-contract deadlines arrive at different calendar times for firms with different fiscal year ends.

$$\text{Calendar seasonality} \perp \text{Fiscal-quarter status conditional on fiscal-year-end variation.}$$

Interpretation: The design is a seasonal difference-in-differences. It compares first/fourth fiscal quarters to other fiscal quarters after absorbing calendar-month patterns.

5.2 Why industry calendar controls are crucial

The paper controls for calendar effects interacted with industry. This permits demand seasonality to differ by industry. For example, holiday effects can differ between computers, drugs, and consumer goods.

Interpretation: Without industry-specific calendar controls, a fourth-fiscal-quarter spike could simply be a holiday spike. The strongest specifications allow calendar seasonality to vary by industry.

5.3 Main threats

- **Endogenous fiscal year choice.** Firms may choose fiscal years to coincide with natural demand cycles.
- **Accounting/audit timing.** Unaudited quarterly statements could generate apparent revenue timing even without real timing changes.
- **Firm-specific demand shocks.** Industries may not be fine enough to absorb all calendar seasonality.
- **Price proxy measurement.** The price measure uses sales minus cost of goods sold, which may mix markups, accounting, and cost shocks.

Interpretation: The paper addresses these threats with finer calendar controls, industry heterogeneity, price evidence, and fiscal-year changes after mergers or rare voluntary changes.

5.4 Validation tests

- Table VI shows fiscal revenue effects remain stable under different calendar controls.
- Table VII examines exogenous fiscal-year changes from mergers.

- Table VIII examines firms that changed fiscal year ends.
- Price tables show a pattern opposite to revenue, consistent with year-end price discounting or effort rather than pure demand timing.

Interpretation: These tests are essential because the paper cannot directly observe the incentive contract of each salesperson. The empirical case is built from multiple indirect implications of nonlinear incentives.

6 Tables and figures

6.1 Figure I: Fiscal-Year Seasonality; Figure II: Sample Compensation Plans

Read: Figure I compares pairs of direct competitors with different fiscal year ends. Computer manufacturers show visible fourth-fiscal-quarter sales spikes, while consumer goods firms show much less fiscal-year timing. Figure II shows nonlinear CEO and salesperson compensation schedules.

Economic message: The two figures motivate the paper. Figure I shows the empirical phenomenon; Figure II shows the incentive mechanism that can produce it.

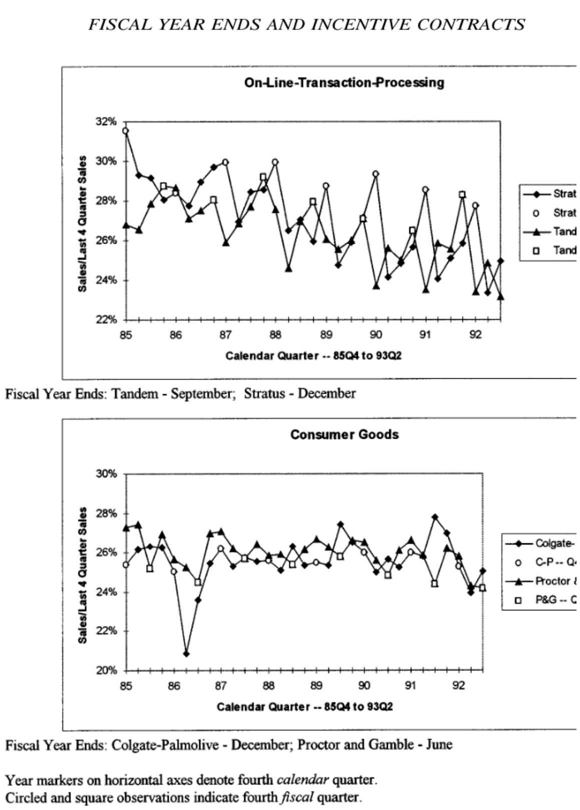

Figure I: Fiscal-Year Seasonality.

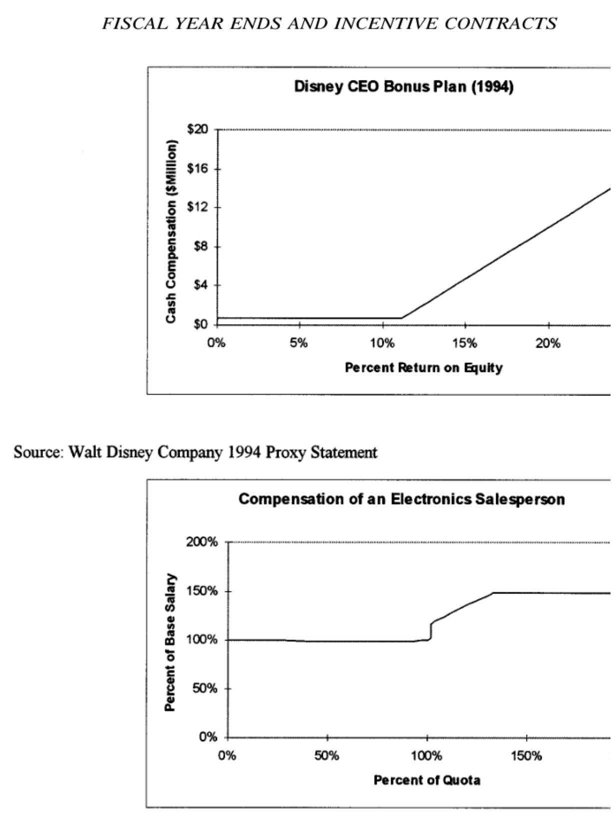

Figure II: Sample Compensation Plans.

The figures pair the motivating evidence and the incentive mechanism. Fiscal-year spikes are more visible when sales timing is plausibly controllable, and nonlinear contracts give agents reason to manipulate timing.

6.2 Figure III: "Timing Gaming" in a Two-Year, Two-Fiscal-Period Model; Table I: Fiscal Year Ends for Selected Three-Digit Industries

Read: Figure III shows pull-in, push-out, and no-gaming regions in the timing model. Table I documents the distribution of fiscal year ends across the sample and selected three-digit industries.

Economic message: The figure provides the model intuition; the table provides the variation needed for identification.

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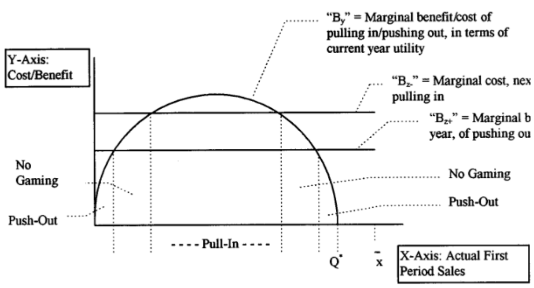

FIGURE III
"Timing Gaming" in a Two-Year Two-Fiscal-Period Model

Figure III: timing-gaming model.

FISCAL YEAR ENDS AND INCENTIVE CONTRACTS

TABLE I
FISCAL YEAR ENDS FOR SELECTED THREE-DIGIT INDUSTRIES

	All	Paper (160)	Drugs (181)	Industrial chemicals (192)	Electric machinery (342)	Scien instru (37
January	13	0	0	0	4	
February	21	1	0	1	2	
March	44	0	8	0	4	
April	14	0	0	0	2	
May	21	0	0	0	1	
June	86	1	2	1	11	
July	26	0	1	0	2	
August	12	0	1	0	3	
September	71	1	3	4	6	
October	30	1	1	0	1	
November	19	0	1	0	0	
December	624	29	31	33	36	2
Total	981	33	48	39	72	5

Table I: fiscal year ends across industries.

Figure III predicts fiscal-year timing manipulation, while Table I shows that fiscal-year-end variation is rich enough to estimate fiscal effects separately from calendar effects.

6.3 Table IIa: Effect of Calendar and Fiscal Year on Revenue; Table IIb: Fiscal Quarter Revenue Effects by Industry

Read: Table IIa shows a first-fiscal-quarter revenue decrease of about 4.8 percent and a fourth-fiscal-quarter increase of about 2.7 percent in the basic specification. Table IIb shows large industry heterogeneity.

Economic message: The main revenue result is that firms sell less early in the fiscal year and more at year end, after calendar controls.

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TABLE IIa

EFFECT OF CALENDAR AND FISCAL YEAR ON REVENUE
DEPENDENT VARIABLE: ln (FISCAL QUARTER SALES) − ln (PREVIOUS FISCAL QUARTER SALES)

	(1)	(2)	(3)
Δ(Fiscal quarter = 1)	---	−.0481 (.0030)	---
Δ(Fiscal quarter = 4)	---	0.0267 (.0030)	---
Δ(Fiscal quarter effects by three-digit industry)	no	no	Reported in Table IIb
Δ(Calendar month effects by three-digit industry)	yes	yes	yes
N (company quarters)	19,732	19,732	19,732
R^2	.141	.167	.191

The sample consists of all available manufacturing firms. Three-digit industry Census classifications were determined from Compustat four-digit SIC codes. All firms in a given three-digit industry were dropped from the sample if the industry failed to meet either of two criteria: 1) not more than 80 percent of the firms in the industry have December as their fiscal year ends, and 2) the sample contains at least five firms in the industry. Each regression includes changes in indicator variables for the interaction of each calender month and each three-digit industry. Column (2) includes change in indicator variables for first fiscal quarter and fourth fiscal quarter. Column (3) includes change in indicator variables of the interaction of each three-digit

Table IIa: average revenue fiscal effects.

TABLE IIb

FISCAL QUARTER REVENUE EFFECTS BY INDUSTRY

Three-digit industry	Q1 effect	Q4 effect
Meat (100)	−.0062 (.0123)	.0293 (.0135)
Canned food (102)	−.1274 (.0255)	−.0437 (.0253)
Grain (110)	−.0035 (.0179)	−.0093 (.0174)
Beverages (120)	−.0517 (.0261)	.0151 (.0287)
Other food (122)	−.1006 (.0350)	−.0556 (.0350)
Knitting (132)	−.1049 (.0293)	.0341 (.0307)
Yarn, thread (142)	−.0546 (.0184)	.0541 (.0209)
Apparel (151)	−.0212 (.0161)	−.0183 (.0188)
Printing (172)	−.0802 (.0129)	−.0151 (.0119)
Drugs (181)	−.0160 (.0141)	−.0015 (.0125)
Soap (182)	−.0166 (.0264)	.0147 (.0198)
Paint (190)	.0343 (.0341)	.1345 (.0390)
Agricultural chemicals (191)	−.0200 (.0574)	.0446 (.0623)
Footwear (221)	−.2817 (.0422)	.1413 (.0447)
Wood homes (232)	−.0702 (.0360)	.0313 (.0341)
Furniture (242)	−.1017 (.0187)	.0411 (.0186)
Cement, etc. (251)	−.1736 (.0787)	−.1173 (.0494)
Steelworks (270)	−.0022 (.0116)	.0202 (.0113)
Fabricated metal (282)	−.0080 (.0240)	.0711 (.0201)
Screw machines (290)	.0210 (.0537)	.0911 (.0608)
Metal forgings (291)	−.0001 (.0177)	.0231 (.0169)
Miscellaneous fabricated metal (300)	−.0307 (.0141)	−.0125 (.0139)
Engines (310)	−.2017 (.0303)	.0055 (.0249)
Farm machinery (311)	−.2667 (.0495)	.0773 (.0471)
Construction machinery (312)	−.0623 (.0252)	.0585 (.0248)
Office machines (321)	−.0441 (.0150)	.0431 (.0155)
Computers (322)	−.0640 (.0173)	.0528 (.0176)
Machinery (331)	−.0567 (.0093)	.0285 (.0112)
Radio, TV (341)	−.0314 (.0124)	.0162 (.0145)
Electronic machinery (342)	−.0233 (.0067)	.0093 (.0079)
Motor vehicles (351)	.0041 (.0150)	.0256 (.0184)
Aircraft (352)	−.0524 (.0208)	.0002 (.0218)
Missiles (362)	−.0575 (.0290)	.0907 (.0269)
Science instruments (371)	−.0431 (.0092)	.0483 (.0099)
Optical supplies (372)	−.0410 (.0128)	.0620 (.0126)
Photo equipment (380)	−.0757 (.0294)	.0423 (.0288)
Miscellaneous manufacturing (391)	−.1081 (.0276)	.0168 (.0320)

Table IIb: industry revenue effects.

The paired tables give both the average effect and the heterogeneous industry pattern. The first-quarter coefficient is negative; the fourth-quarter coefficient is positive.

6.4 Table IIIa: Effect of Calendar and Fiscal Year on Prices; Table IIIb: Fiscal Quarter Price Effects by Industry

Read: Table IIIa shows that prices rise in the first fiscal quarter and fall in the fourth fiscal quarter. Table IIIb shows the price pattern varies by industry.

Economic message: The price pattern is the mirror image of the revenue pattern and supports the idea that year-end revenue is boosted partly through pricing or effort.

TABLE IIIa
EFFECT OF CALENDAR AND FISCAL YEAR ON PRICES
DEPENDENT VARIABLE: CHANGE IN (ln(SALES) − ln(COST OF GOODS SOLD))
FROM PREVIOUS QUARTER

	(1)	(2)	(3)
Δ(Fiscal quarter = 1)	---	0.0072 (.0018)	---
Δ(Fiscal quarter = 4)	---	−.0166 (.0019)	---
Δ(Fiscal quarter effects by industry)	no	no	Reported in Table IIIb
Δ(Calendar month effects by industry)	yes	yes	yes
N (company quarters)	18,792	18,792	18,792
R^2	.030	.039	.048

The dependent variable is the quarterly change in the log sales minus log cost of goods sold. If direct costs per unit of output are orthogonal to the fiscal year end, then the fiscal year variables in this table approximate the percentage by which average prices in the first or fourth fiscal quarter differ from the second and third fiscal quarters, while controlling for calendar effects on cost and price. The sample, sample restrictions, industry classifications, and calendar month controls are the same as in Table IIa. There are fewer

Table IIIa: average price fiscal effects.

TABLE IIIb
FISCAL QUARTER PRICE EFFECTS BY INDUSTRY

Three-digit industry	Q1 effect	Q4 effe
Meat (100)	.0108 (.0087)	−.0078 (.0
Canned food (102)	.0279 (.0135)	−.0032 (.0
Grain (110)	.0112 (.0112)	−.0052 (.0
Beverages (120)	.0739 (.0284)	.0347 (.0
Other food (122)	−.0184 (.0112)	−.0038 (.0
Knitting (132)	.0006 (.0069)	.0024 (.0
Yarn, thread (142)	.0059 (.0102)	.0065 (.0
Apparel (151)	.0133 (.0104)	−.0160 (.0
Printing (172)	−.0036 (.0058)	−.0185 (.0
Drugs (181)	.0011 (.0148)	−.0184 (.0
Soap (182)	.0003 (.0135)	−.0672 (.0
Paint (190)	.0173 (.0098)	−.0124 (.0
Agricultural chemicals (191)	.0595 (.0216)	−.0722 (.0
Footwear (221)	.0055 (.0110)	.0354 (.0
Wood homes (232)	−.0033 (.0059)	−0089 (.0
Furniture (242)	−.0004 (.0098)	−0059 (.0
Cement, etc. (251)	−.0543 (.0237)	−.0358 (.0
Steelworks (270)	.0044 (.0052)	.0021 (.0
Fabricated metal (282)	.0067 (.0099)	−.0086 (.0
Screw machines (290)	.0341 (.0257)	−.0437 (.0
Metal forgings (291)	.0000 (.0081)	−.0084 (.0
Miscellaneous fabricated metal (300)	−.0011 (.0128)	−.0223 (.0
Engines (310)	−.0212 (.0124)	−.0311 (.0
Farm machinery (311)	−.0041 (.0170)	−.0298 (.0
Construction machinery (312)	.0372 (.0120)	−.0195 (.0
Office machines (321)	−.0006 (.0111)	−.0490 (.0
Computers (322)	.0169 (.0083)	−.0176 (.0
Machinery (331)	.0033 (.0037)	−.0057 (.0
Radio, TV (341)	.0016 (.0048)	−.0140 (.0
Electronic machinery (342)	.0224 (.0062)	−.0263 (.0
Motor vehicles (351)	.0006 (.0040)	.0012 (.0
Aircraft (352)	.0059 (.0089)	−.0075 (.0
Missiles (362)	−.0016 (.0060)	−.0085 (.0
Science instruments (371)	.0003 (.0051)	−.0269 (.0
Optical supplies (372)	.0184 (.0102)	−.0088 (.0
Photo equipment (380)	−.0328 (.0359)	−.1004 (.0
Miscellaneous manufacturing (391)	.0042 (.0109)	.0076 (.0

Table IIIb: industry price effects.

The price evidence is important because it indicates that the sales pattern is not purely a demand timing story. If prices are lower in the fourth fiscal quarter, agents may be discounting or exerting effort to pull sales forward.

6.5 Table IV: Determinant of Interindustry Variation in Fiscal-Revenue Effects; Table V: Determinants of Interindustry Variation in Fiscal-Price Effects

Read: Table IV relates revenue fiscal effects to product and sales-force characteristics. Direct and wholesale channels and turnover help explain variation. Table V performs the analogous analysis for prices.

Economic message: Industry heterogeneity supports the incentive-contract interpretation: fiscal effects are stronger where salespeople can influence timing and where turnover changes incentives.

TABLE IV

DETERMINANT OF INTERINDUSTRY VARIATION IN FISCAL-REVENUE EFFECTS
DEPENDENT VARIABLE: INDUSTRY FISCAL-REVENUE EFFECTS

	(1)	(2)	(3)		(4)	(5)	(6)
	By four-digit SIC code				By three-digit Census industry (see Table IIb)		
FQ = 1:				FQ = 1:			
Durable	.0136		−.0045	Durable	−.0127		
	(.0137)		(.0160)		(.0199)		
Direct		.0488		Direct	.0452		
		(.0246)			(.0352)		
Wholesale			−.0736	Turnover	−.0137	−.0252	
			(.0431)		(.0718)	(.0733)	
Retail			−.0818	Education			.0166
			(.0472)				(.0185)
Other nonmfg			−.0156	Earnings			−.1243
			(.0384)				(.1147)
Government			−.0634				
			(.0410)				
FQ = 4:				FQ = 4:			
Durable	.0363		.0371	Durable	.0211		
	(.0141)		(.0167)		(.0200)		
Direct		.0542		Direct	.0298		
		(.0241)			(.0361)		
Wholesale			−.0076	Turnover	.1016	.0814	
			(.0448)		(.0737)	(.0744)	
Retail			.0335	Education			−.0191
			(.0473)				(.0177)
Other nonmfg		.0327	.1016	Earnings			.1481
		(.0399)	(.0400)				(.1071)
Government			.0327				
			(.0399)				
N	102	102	102		74	74	74
R^2	.466	.459	.541		.552	.505	.535

An observation is a manufacturing industry. The dependent variable is the industry fiscal-revenue effect for the first or fourth fiscal quarter. The three-digit coefficients (columns (4)–(6)) are detailed in Table IIb. Each explanatory variable is interacted with indicator variables for the first or fourth fiscal quarters. (The regressions cannot be run separately for each quarter's effects while using Hanushek's [1974] methodology.) Classification as durable or nondurable is based on Compustat four-digit SIC code and Census Bureau definitions of durable industries. Data for customer channels are from the 1987 Census of Manufacturers

Table IV: determinants of revenue effects.

TABLE V

DETERMINANTS OF INTERINDUSTRY VARIATION IN FISCAL-PRICE EFFECTS
DEPENDENT VARIABLE: INDUSTRY FISCAL-PRICE EFFECTS

	(1)	(2)	(3)		(4)	(5)	(
	By four-digit SIC code				By three-digit Censu industry (see Table II		
FQ = 1:				FQ = 1:			
Durable	−.0053		−.0047	Durable	−.0042		
	(.0046)		(.0056)		(.0054)		
Direct		−.0002		Direct	.0050		
		(.0071)			(.0091)		
Wholesale			−.0012	Turnover	.0102	.0073	
			(.0147)		(.0168)	(.0153)	
Retail			−.0002	Education			−.0
			(.0119)				(.0
Other nonmfg			.0007	Earnings			.0
			(.0143)				(.0
Government			.0018				
			(.0106)				
FQ = 4:				FQ = 4:			
Durable	.0025		.0100	Durable	.0028		
	(.0049)		(.0066)		(.0070)		
Direct		−.0087		Direct	−.0127		
		(.0077)			(.0117)		
Wholesale			.0054	Turnover	−.0198	−.0170	
			(.0168)		(.0190)	(.0183)	
Retail			.0180	Education			−.0
			(.0133)				(.0
Other nonmfg			−.0211	Earnings			.0
			(.0195)				(.0
Government			−.0036				
			(.0115)				
N	102	102	102		74	74	74
R^2	.157	.151	.205		.291	.261	.3

An observation is a manufacturing industry. The dependent variable is the industry fiscal-price e the first or fourth fiscal quarter. The three-digit coefficients (columns (4)–(6)) are detailed in Table II explanatory variable is interacted with indicator variables for the first or fourth fiscal quarte regressions cannot be run separately for each quarter's effects while using Hanushek's [1974] meth

Table V: determinants of price effects.

The second-stage tables connect the fiscal-quarter coefficients to industry-level mechanisms. The strongest interpretation comes from consistency across revenue and price channels.

6.6 Table VI: Fiscal Quarter Revenue Effects with Differing Calendar Controls; Table VII: Reaction of Seasonality to Exogenous Fiscal-Year Changes; Table VIII: Effect of Fiscal-Year-End Changes

Read: Table VI shows that fiscal effects are robust to varying calendar controls. Table VII uses mergers that change fiscal years, and Table VIII studies firms that change fiscal year ends.

Economic message: These tables address the main alternative explanation: firms might choose fiscal years to match natural seasonality. The robustness and fiscal-year-change evidence support a causal fiscal-year timing effect.

FISCAL YEAR ENDS AND INCENTIVE CONTRACTS

TABLE VI

FISCAL QUARTER REVENUE EFFECTS WITH DIFFERING CALENDAR CONTRO

DEPENDENT VARIABLE: ln(FISCAL QUARTER SALES) − ln(PREVIOUS FISCAL QUARTER SALES)

	(1)	(2)	(3)	(4)	(5)	
Δ(Fiscal quarter = 1)	−.0468	−.0380	−.0411	−.0435	−.0477	−.
	(.0022)	(.0030)	(.0034)	(.0035)	(.0037)	(.
Δ(Fiscal quarter = 4)	.0233	.0332	.0289	.0322	.0271	.
	(.0023)	(.0030)	(.0035)	(.0035)	(.0037)	(.
Δ(Calendar month effects)	none	none	yes	interacted with two-digit SIC	interacted with three-digit Census classification	inte w fc di S
N (company quarters)	23,708	12,240	12,240	12,240	12,240	12,
R^2	0.038	.040	.051	.180	.220	.

See notes to Table IIa for details about the regressions. In all columns the first two rows

Table VI: calendar-control robustness.

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TABLE VII

REACTION OF SEASONALITY TO EXOGENOUS FISCAL-YEAR CHANGES

DEPENDENT VARIABLE: ln(FISCAL QUARTER SALES) − ln(PREVIOUS FISCAL QUARTER SALES)

	(1) Unisys	(2) Philip Morris
Δ(Calendar quarter = 1) interacted with postmerger indicator	−.1644	0.0303
	(.0263)	(.0147)
Δ(Calendar quarter = 2) interacted with postmerger indicator	0.0040	0.0293
	(.0226)	(.0127)
Δ(Calendar quarter = 4) interacted with postmerger indicator	0.0871	0.0490
	(.0226)	(.0131)
N (quarters)	51	51
R^2	.893	.678

In 1986, Sperry (March fiscal year end) and Burroughs (December fiscal year end) merged to form Unisys (December fiscal year end). In 1986, General Foods (March fiscal year end) was purchased by Philip Morris (December fiscal year end). The merged company has a December fiscal year end. An observation is sales of the company during a postmerger calendar quarter or combined sales of the two companies during a premerger calendar quarter. The omitted category is the third calendar quarter, because it is never the first or fourth fiscal quarter of any of the companies. These regressions also include change in first, second, and fourth calendar indicators as explanatory variables. The data set includes fiscal years 1980–1993.

Table VII: merger-induced fiscal-year changes.

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TABLE VIII

EFFECT OF FISCAL-YEAR-END CHANGES

DEPENDENT VARIABLE: ln(FISCAL QUARTER SALES) − ln(PREVIOUS FISCAL QUARTER SALES)

Δ(Fiscal quarter = 1)	−0.0094
	(.0154)
Δ(Fiscal quarter = 4)	0.0311
	(.0151)
Δ(Calendar quarter effects by company)	yes
N (company quarters)	309
R^2	.711

The sample consists of American-based manufacturing firms that changed their fiscal year ends. All included companies met the following criteria: 1) at least two years of quarterly data were available before and after the change in fiscal year; 2) the fiscal year changed by a multiple of three months, 3) quarterly sales for all included quarters were at least $2.5 million, 4) net worth was positive at the time the fiscal year changed; and 5) log sales did not increase by more than one or decrease by more than 0.5 in the year following the fiscal-year change. Sales are deflated by the CPI. Standard errors (in parentheses) are corrected for heteroskedasticity and for serial correlation.

Table VIII: fiscal-year-end changes.

The three robustness tables show that the revenue pattern is not simply calendar seasonality or endogenous fiscal-year selection.

6.7 Appendix 1: Sample industries

Read: Appendix 1 summarizes the industry sample used for the fiscal-year analysis, including the number of companies, observed quarters, December fiscal-year share, and average quarterly sales.

Economic message: This sample table clarifies which industries provide the variation behind the main estimates.

APPENDIX 1:
(CONTINUED)

Three-digit Census industry	Number of companies	Number of observed quarters	Fraction with December fiscal year end	Ave quar sa ($mi
Printing (172)	29	940	0.65	2
Plastics (180)	9	280	1	13
Drugs (181)	48	1420	0.66	4
Soap (182)	17	596	0.58	5
Paint (190)	7	252	0.71	3
Agric chem (191)	7	148	0.51	4
Indus chem (192)	39	1272	0.85	7
Petroleum (200)	37	1224	0.91	42
Coal (201)	2	72	0.5	
Tires (210)	3	108	1	10
Other rubber (211)	3	104	0.35	4
Misc plastic (212)	27	880	0.80	1
Footwear (221)	9	316	0.32	
Leather (222)	2	72	0.50	
Sawmills (231)	9	248	1	9
Wood homes (232)	6	204	0.18	
Furniture (242)	12	400	0.36	1
Glass (250)	3	100	0.72	5
Cement, etc. (251)	9	324	0.78	1
Pottery (261)	1	16	1	
Misc nonmetal (262)	2	72	1	18
Steelworks (270)	29	920	0.53	3
Iron (271)	2	72	0.50	1
Aluminum (272)	23	728	1	4
Cutlery (281)	10	360	0.80	2
Fab metal (282)	8	240	0.58	
Screw mach (290)	6	204	0.67	4
Metal forgings (291)	8	248	0.60	
Ordnance (292)	3	88	0.77	
Misc fab met (300)	20	680	0.69	1
Engines (310)	6	216	0.50	9
Farm mach (311)	10	288	0.39	10
Constr mach (312)	15	520	0.68	3
Metalwork (320)	9	312	0.83	1
Office mach (321)	21	656	0.48	1
Computers (322)	20	692	0.54	14
Machinery (331)	47	1588	0.56	1
Appliances (340)	9	244	1	4
Radio, TV (341)	30	964	0.50	4
Elec mach (342)	72	2364	0.49	6
Motor veh (351)	42	1328	0.74	29

Appendix 1: industry sample characteristics.

7 Short summary

- The paper studies how nonlinear compensation contracts affect business seasonality.
- Fiscal-year-end variation permits separation of fiscal-year seasonality from calendar seasonality.
- Sales are lower in the first fiscal quarter and higher in the fourth fiscal quarter.
- Prices move in the opposite direction: higher early and lower at year end.
- The fiscal-year effects are stronger in industries where salespeople or managers can influence customer purchase timing.
- The results survive several alternative-explanation tests, including differing calendar controls and fiscal-year changes.
- The paper links micro incentive contracts to aggregate-looking seasonal patterns in sales and prices.

8 Conclusion

8.1 Core conclusion

Oyer shows that fiscal-year incentives affect real economic behaviour. Nonlinear compensation contracts create incentives for salespeople and executives to pull sales into the end of the fiscal year and alter prices or effort around fiscal-year boundaries.

8.2 Economic intuition

A quota or bonus threshold changes the marginal value of a sale. Near year end, an extra sale can determine whether an agent earns a bonus. This makes the timing of sales valuable to the agent even if it is not valuable to the firm.

8.3 Contribution revisited

The paper contributes by showing that agency problems appear in product-market timing, not only in accounting choices or executive pay levels. It also provides a clever identification strategy: fiscal-year variation separates incentive timing from calendar demand seasonality.

8.4 Policy and organizational implications

The paper suggests that firms should not view fiscal-year quotas or bonus thresholds as neutral. Such contracts may distort pricing, production smoothing, customer timing, and sales effort. Better-designed contracts may need rolling targets, smoother payout schedules, or mechanisms that reduce end-of-year manipulation incentives.

8.5 Limitations

The data do not directly observe every salesperson's contract or every timing decision. The causal case is built from indirect evidence: fiscal-year timing, industry heterogeneity, price movements, and fiscal-year changes. The paper is strongest as a broad demonstration of incentive-induced timing effects rather than a precise estimate of welfare losses.

8.6 Takeaway

The central takeaway is that compensation calendars can shape business calendars. When agents are rewarded over fiscal-year windows, fiscal years become real economic deadlines that influence sales, prices, and production timing.